

GURU NANAK DEV ENGINEERING COLLEGE, LUDHIANA

Department of CSE

ASSIGNMENT PLAN

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Branch	Semester	Subject	Code	Session
CSE	3rd	Data Structures	CCS203	July-Dec.2025

Assignment No.-1

Content Covered:

Unit-1 Basic concepts and Arrays- Concept of data type, Linear and non-linear data structures, Data structures versus data types, Operations on data structures, Algorithm complexity and Asymptotic notations. Linear and multidimensional arrays and their representation, Operations on arrays, Sparse matrices and their storage.

Unit-2 Searching and Sorting- Linear and binary search techniques, Sorting methods – Bubble sort, Selection sort, Insertion sort, Merge sort, Shell sort and radix sort. Complexities of searching and sorting algorithms.

Unit -3 Linked List- Definition and representation of Linked list, Types of Linked list- Linear linked list, Doubly linked list, Circular linked list and Header linked list and their operations, Application of linked lists.

Unit -4 Stacks- Sequential and Linked representation of stacks, Operations on stacks, Application of stacks – parenthesis checker, Evaluation of postfix expressions, Conversion from infix to postfix, Conversion from infix to prefix representation, Tower of Hanoi problem, implementing recursive functions, Quick sort.

Q1 (a) Find upper bound, lower bound and tight bound for the following function

$$10n^2 + 4n + 2$$

(b) What is Two-dimensional array? How it is represented in memory?

Q2. A sorted list of roll numbers is given:

[101, 104, 110, 115, 120, 125, 130, 140, 150, 160]

The school wants to quickly check whether a student with roll number 125 is present.

1. Show how binary search will trace step by step to find 125.
2. How many comparisons are needed?
3. If the roll number 135 is searched (which is not in the list), explain step by step how binary search will decide that it does not exist.
4. Why is binary search more efficient than linear search in this case?

Q3. Given an integer K, write a procedure which deletes the Kth element from a linked list.

Q4. Convert following infix expression to an equivalent postfix expression: $A+B*C-D/F$

If values of $A=5$, $B=8$, $C=6$, $D=10$ and $E=2$ evaluate the postfix expression using stacks.